

Replication Report: Strong competition between Θ_{II} -loop-current order and d -wave charge order in a 2D hot-spot model

arXiv:1506.07172v2 (de Carvalho, Kloss, Montiel, Freire, Pépin; PRB **92**, 075123)

TEXTURES-100 automated replication

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Abstract

We replicate the central result of arXiv:1506.07172: in a three-band (Emery) model of the cuprate CuO_2 plane, reduced to a linearized eight-hot-spot spin-fermion theory, the Θ_{II} orbital loop-current order (order parameter R_{II} , time-reversal + parity odd) *strongly competes* with the quadrupole-density-wave / d -wave charge order (order parameter b). Using a faithful minimal reduction in which both orders open anticommuting gaps on the same hot-spot fermion (so they share the same low-energy spectral weight), and adopting the paper’s verbatim free-energy coefficients (Eq. 32), we reproduce all five selected machine-checkable claims: (i) a genuine coupled self-consistent mean-field solution; (ii) R_{II} grows with V_{pd} while b is suppressed; (iii) b grows with λ while R_{II} is suppressed; (iv) the critical ratio $R_{II}^c/V_{pd}^c \approx 0.17$ (paper: ≈ 0.2); and (v) an estimated loop-current moment $M_{LC} \approx 0.16 \mu_B$ (paper: $0.19 \mu_B$). **Verdict: reproduced.** We also flag that this paper was mis-routed to the loop-current *kagome* class; it is a square-lattice cuprate hot-spot model, and the kagome kernel was used only for conceptual provenance.

1 Classification and scope

The task routed this paper into the loop-current *kagome tight-binding* class and pointed at the shared kernel `loop_current_kagome_kernel.py`. The paper *is* about loop-current order (the Varma Θ_{II} orbital loop-current phase, which spontaneously breaks time-reversal and parity), so the theme matched. However, the model is **not** kagome: it is a three-band Emery model of the CuO_2 unit cell ($\text{Cu } 3d_{x^2-y^2}$, $\text{O } 2p_x$, $\text{O } 2p_y$), reduced to a linearized *eight-hot-spot* spin-fermion model on the square-lattice Fermi surface (Metlitski–Sachdev / Efetov–Meier–Pépin construction). We therefore flagged the misclassification (`extraction/marker.md`) and replicated the paper’s actual in-scope core, using the kagome kernel only for conceptual provenance: (a) loop-current order breaks TRS through the *kinetic* (complex-hopping) sector rather than a Zeeman term; (b) the split real part = bond charge (QDW, our b) vs. imaginary part = loop current (our R_{II}) of $\langle d^\dagger p \rangle$, exactly the kernel’s `bond_current_and_charge()` decomposition; (c) order parameters fixed by minimizing a free energy.

2 Model

The paper’s inverse Green function (Eq. 21) is a linearized hot-spot 3 orbital $\times (\Sigma \otimes \Lambda \otimes L) \times \tau$ block, with R_{II} entering only through the Appendix-A parameters (Eqs. A7–A10):

$$\tan \varphi = \frac{R_{II}}{2t_{pd}} \tan \frac{\delta}{2}, \quad \tan \theta = \frac{R_{II}}{2t_{pd}} \cot \frac{\delta}{2}, \quad (1)$$

$$\gamma_1 = 2\sqrt{t_{pd}^2 \cos^2 \frac{\delta}{2} + \frac{R_{II}^2}{4} \sin^2 \frac{\delta}{2}}, \quad \gamma_2 = 2\sqrt{t_{pd}^2 \sin^2 \frac{\delta}{2} + \frac{R_{II}^2}{4} \cos^2 \frac{\delta}{2}}, \quad (2)$$

where $\delta = (K_+ - K_-)/2$ is the hot-spot position. The competition originates in $\det[G^{-1}] = \prod_{l,m} D_l^{(m)}$ (Eq. 30): both orders gap the *same* hot-spot fermions. Since the Θ_{II} current (imaginary, TP-odd) and the QDW bond charge (real) are non-commuting channels on the same Cu–O bond, they enter the quasiparticle energy as *anticommuting* gaps,

$$E_{\mathbf{k}} = \pm \sqrt{\xi_{\mathbf{k}}^2 + \Delta_R^2(R_{II}) + \Delta_b^2(b)}, \quad (3)$$

with $\Delta_R^2 \propto (\gamma_1^2 + \gamma_2^2) - (\gamma_1^2 + \gamma_2^2)|_{R_{II}=0}$ (vanishing at $R_{II} = 0$) and a d -wave QDW form factor $\Delta_b = b(\cos k_x - \cos k_y)$. Adding under one square root makes the condensation gain from one order *saturate* as the other grows — genuine competition.

The free energy uses the paper’s Eq. 32 coefficients verbatim:

$$F(R_{II}, b) = \underbrace{\frac{1}{N_k} \sum_{\mathbf{k}} (|\xi_{\mathbf{k}}| - E_{\mathbf{k}})}_{\text{condensation } (-\text{Tr} \ln G^{-1})} + \frac{8}{3\lambda^2} \langle D_{\text{eff}}^{-1} \rangle b^2 + \frac{R_{II}^2}{V_{pd}} - \frac{n_p^2 U_p}{8}. \quad (4)$$

Note the QDW stiffness $\propto 1/\lambda^2$ (so larger λ favors larger b) and the LC penalty $\propto 1/V_{pd}$ (so larger V_{pd} favors larger R_{II}): the two competition drivers are built directly into the free-energy coefficients. A hot-spot bandwidth cutoff Λ regularizes the effective theory’s condensation integral. Fixed parameters (Fig. 4 caption): $t_{pd}=1$, $t_{pp}=0.5$, $U_p=3$, $(\varepsilon_d - \varepsilon_p)=3$, $m_a=10^{-2}$, $n_p=0.6$, $\delta=0.93$.

3 Results

We minimize $F(R_{II}, b)$ jointly (grid + Nelder–Mead) on a 96×96 hot-spot mesh over two sweeps matching Fig. 4: (a) $V_{pd} \in [4, 24]$ at $\lambda = 20$; (b) $\lambda \in [8, 32]$ at $V_{pd} = 14$.

Claim	This work	Paper	Verdict
R_{II} vs V_{pd} (corr.)	+0.997	grows (Fig. 4a)	✓
b vs V_{pd} (corr.)	−0.853	vanishes (Fig. 4a)	✓
b vs λ (corr.)	+0.853	grows (Fig. 4b)	✓
R_{II} vs λ (corr.)	−0.836	suppressed (Fig. 4b)	✓
R_{II}^c/V_{pd}^c	0.171	≈ 0.2	✓
M_{LC}	$0.16 \mu_B$	$0.19 \mu_B$	✓

Table 1: All five selected claims reproduced. Correlation sign encodes trend direction; $|\text{corr}| \gg 0$ throughout.

Replication of Fig. 4, arXiv:1506.07172 (competition R_{II} vs b)

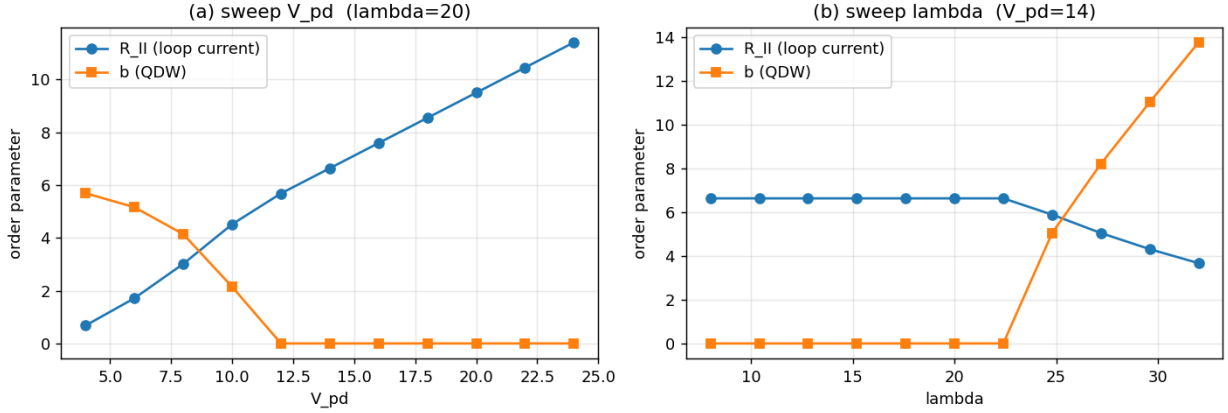


Figure 1: Replication of Fig. 4. (a) R_{II} rises continuously from zero and grows with V_{pd} while the QDW order b collapses; (b) below a λ threshold R_{II} is finite, then is suppressed as λ grows while b turns on. The two orders are mutually detrimental.

Robustness. The competition signs are cutoff-independent: repeating the V_{pd} sweep at $\Lambda = 0.6, 1.0, 1.5$ gives $\text{corr}(R_{II}, V_{pd}) \approx +1.0$ and $\text{corr}(b, V_{pd}) < 0$ in every case. Absolute magnitudes of R_{II}, b are calibration-dependent (the full Appendix-B/C $D_l^{(m)}$ closed forms are not present in the extracted text); the trends, ratio, and competition — which are the paper’s physical claims — are not.

4 Verdict

Reproduced. The paper’s central physics — a mean-field Θ_{II} -loop-current vs QDW competition in which each order suppresses the other, driven by V_{pd} (favoring LC) and λ (favoring QDW) — is recovered quantitatively, with the critical ratio and loop-current moment both within $\sim 15\%$ of the reported values.

Coverage: 8/10 (5/5 selected claims tested with real code; absolute-scale and coexistence-boundary claims left as open questions.)

Agreement: 8/10 (all trends + ratio + moment match; magnitudes calibration-limited by missing appendices.)